Chih-Fan (Rich) Pai

Ph.D. Candidate, UC San Diego

Electrical and Computer Engineering
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Research Interests

My research interests include optimization, online learning, and interactive decision-making, with a focus on control, prediction, and reinforcement learning for dynamical systems.

Education

University of California, San Diego Dec. 2026 (Expected)
Ph.D., Electrical and Computer Engineering
Graduate Scholarly Impact Award; J. Yang Scholarship

National Taiwan University 2020
M.S., Communication Engineering
Best Master's Thesis Award; Youth Thesis First Prize

National Yang Ming Chiao Tung University 2018
B.S., Electrical and Computer Engineering
Academic Excellence Award

Publications

- **C.-F. Pai** and Y. Zheng. *Online Nonstochastic Prediction: Logarithmic Regret via Predictive Online Least Squares*. *Preprint*, 2026.
- **C.-F. Pai**, Y. Watanabe, Y. Tang, and Y. Zheng. *Policy Optimization of Mixed $\mathcal{H}_2/\mathcal{H}_\infty$ Control: Benign Nonconvexity and Global Optimality*. Provisionally accepted at *Automatica*, 2026.
- **C.-F. Pai**, X. Shang, J. Qian, and Y. Zheng. *Online Tracking with Predictions for Nonlinear Systems with Koopman Linear Embedding*. *Learning for Dynamics and Control (L4DC)*, PMLR 331:1575–1600, 2026.
- Y. Zheng, **C.-F. Pai**, and Y. Tang. *Benign Nonconvex Landscapes in Optimal and Robust Control, Part I: Global Optimality*. *IEEE Transactions on Automatic Control*, 71(8):5334–5349, 2026.
- Y. Zheng, **C.-F. Pai**, and Y. Tang. *Benign Nonconvex Landscapes in Optimal and Robust Control, Part II: Extended Convex Lifting*. *IEEE Transactions on Automatic Control*, 71(8):5350–5365, 2026.
- Y. Watanabe, **C.-F. Pai**, and Y. Zheng. *Semidefinite Programming Duality in Infinite-Horizon LQ Differential Games*. *IEEE Conference on Decision and Control (CDC)*, pp. 3964–3970, 2025.
- Y. Zheng, **C.-F. Pai**, and Y. Tang. *Extended Convex Lifting for Policy Optimization of Optimal and Robust Control*. *Learning for Dynamics and Control (L4DC)*, PMLR 283:392–404, 2025.

Research Experience

University of California, San Diego 2023–Present
Ph.D. Research Assistant, Advisor: Professor Yang Zheng

Online Prediction and Control of Dynamical Systems

2025–Present

- Designed and analyzed sequential prediction and online non-stochastic control algorithms in uncertain and nonstationary dynamic environments using tools from online learning and optimization.
- Developed sequential output prediction algorithms for linear dynamical systems with logarithmic regret

guarantees via online least squares with tailored predictive hints.

- Developed data-driven predictive tracking/control algorithms for Koopman linearizable nonlinear systems and derived its dynamic regret guarantees.

Convex and Nonconvex Optimization in Control
2023–2025

- Analyzed the optimization landscapes of classical optimal and robust control problems from a modern policy optimization perspective, including LQR and H-infinity robust control.
- Established a new geometric and structural characterization of mixed $\mathcal{H}_2/\mathcal{H}_\infty$ control, uncovering connections between policy optimization, classical Riccati theory, and convex reformulations.
- Developed a unified Extended Convex Lifting framework that reveals hidden convexity in nonconvex control policy optimization, covering both static state-feedback and dynamic output-feedback policies.

Selected Talks

- **Decision-making in Uncertain Dynamic Environments**
  Invited seminar, CS Department, National Yang Ming Chiao Tung University, Dec. 2025.
- **Online Non-stochastic Control: A Regret-Minimizing Approach**
  Invited presentation, DSC 291: Online Learning
  Host: Professor Yoav Freund, UC San Diego, March 2025.
- **Policy Optimization for Mixed Control: Benign Nonconvexity and Global Optimality**
  Southern California Control Workshop, University of Southern California, Nov. 2024.

Industry Experience

Qualcomm, Audio R&D Intern Summer 2025

- Designed adaptive active noise cancellation algorithms under real-time and hardware constraints.
- Balanced convergence, robustness, and stability using optimization and robust control theory.

Teaching Experience

Teaching Assistant, UC San Diego

ECE 272A: Stochastic Processes and Dynamical Systems
ECE 285: Agentic AI and LLMs for Smart Grids
ECE 228: Machine Learning for Physical Applications
ECE 285: Semidefinite and Sum-of-Squares Optimization
ECE 171A: Linear Control System Theory
ECE 101: Linear Systems Fundamentals

- Led discussion sessions and developed instructional materials.
- Received consistently strong student evaluations.

Teaching Assistant, National Taiwan University

Linear Algebra (EE), Calculus (MATH), Digital and Multi-rate Signal Processing (CommE)